

Level 1 / Semester II

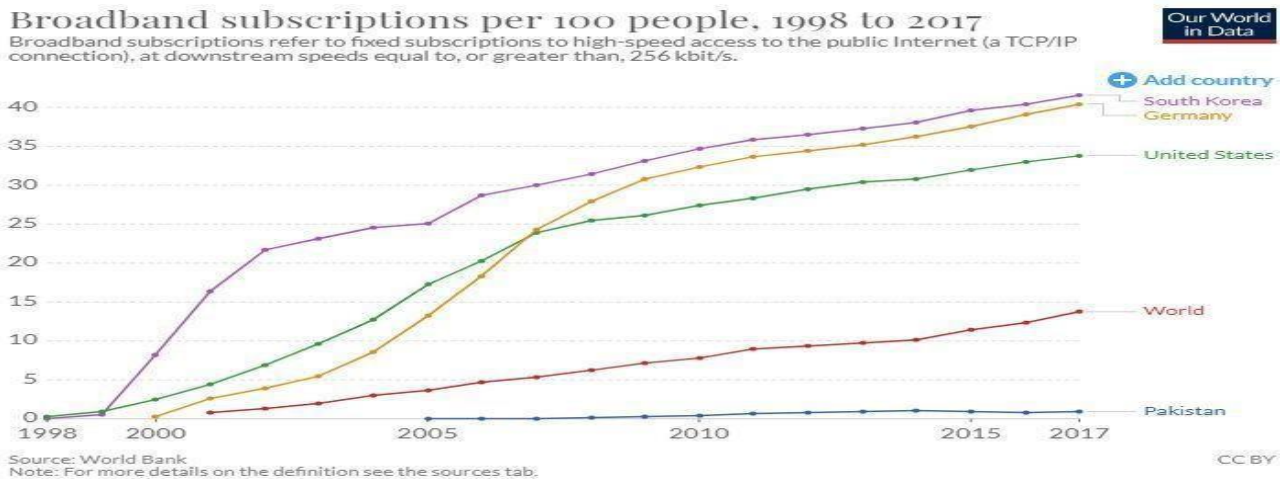
Lesson 3

Broadband Connections

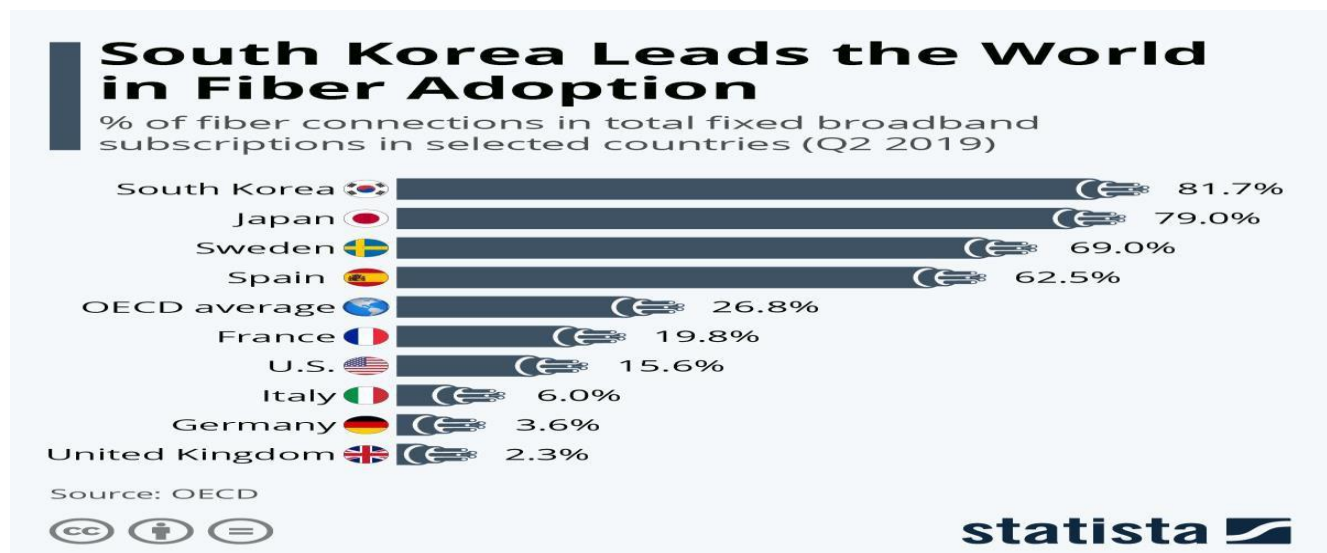
Starter

Study these two graphs and decide if the statements are correct or wrong. Correct the wrong ones.

A



B

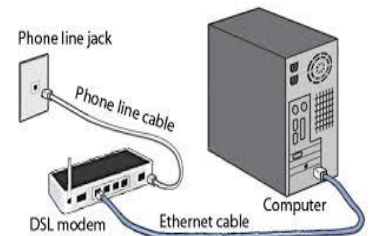


- Broadband subscriptions have increased all over the world except for Pakistan.
- South Koreans account for the second highest rate of broadband subscriptions per 100 people.
- There is a significant digital gap between South Korea and Germany.
- Europe is the leading continent in terms of fiber adoption.
- Fiber cable is used in approximately 20% of connections in France.
- Both France and Spain share equal rates of fiber connections.
- Sweden goes beyond Japan by 10% in terms of broadband subscriptions.

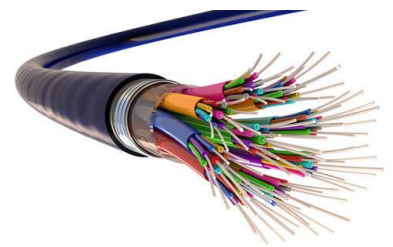
READING

Broadband is a high-speed internet connection that allows you to surf the internet at top speeds. It is one of the most popular choices of internet connection as it's extremely quick and enables us to make the most of the internet. The most common internet mediums include copper phone lines and optical fibre. In technical terms, broadband is a wide bandwidth data transmission where all sorts of traffic and signals are transported and interchanged, providing top speeds at the click of a mouse. Before broadband was around, the internet was only accessible through several dial-up connections. Dial-up is the only non-broadband service available for the internet, but it's very slow by today's standards and is generally no longer a popular option. There are many types of broadband connections available today. These are the most important ones.

DSL (Digital Subscriber Line) was one of the earliest forms of broadband to be introduced and is still a common service today. DSL uses a single pair of copper wires in a telephone line for high-speed data connection. The DSL signal uses different frequencies from the voice lines, making it possible to use your internet and phone at the same time. Using a modem to bring the connection from your telephone line to your computer allows for DSL to get more "bandwidth," which increases speeds. DSL speeds can be as high as 100 Mbps.



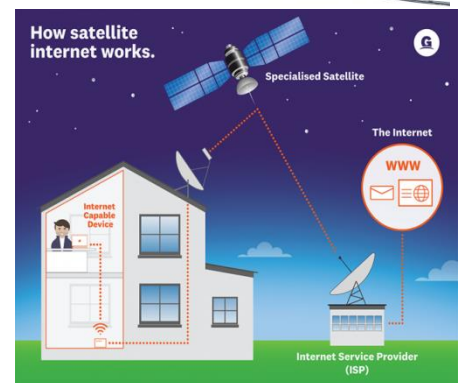
Fiber internet uses a special type of fiber-optic cable, which allows data to be sent as pulses of light over thousands of tiny, transparent fiber strands. Because light is the fastest way to transmit data, fiber-optic internet is incredibly fast, and results in a high-quality and reliable connection. It can provide connections of up to 1 Gbps, and this is expected to get even higher over time. Fiber is an expanding technology that will soon become more widely available.



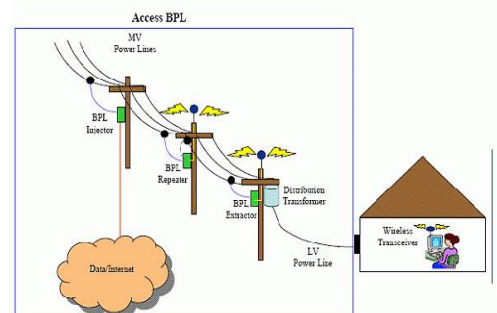
Cable internet is about as common as DSL. It uses the same coaxial cables that transmits TV service to carry a high-speed internet connection. Using a modem, you can connect to these cable lines and enjoy a high-speed, uninterrupted data connection. Cable internet speeds can be up to 940 Mbps for downloading and up to 50 Mbps for uploading.



Satellite broadband is another form of wireless broadband, and is also useful for serving remote or sparsely populated areas. Downstream and upstream speeds for satellite broadband depend on several factors, including the provider and service package purchased, the consumer's line of sight to the orbiting satellite, and the weather. Typically a consumer can expect to receive (download) at a speed of about 500 Kbps and send (upload) at a speed of about 80 Kbps. These speeds may be slower than DSL and cable modem, but they are about 10 times faster than the download speed with dial-up Internet access. Service can be disrupted in extreme weather conditions.



BPL is the delivery of broadband over the existing low- and medium-voltage electric power distribution network. BPL speeds are comparable to DSL and cable modem speeds. BPL can be provided to homes using existing electrical connections and outlets. BPL is an emerging technology that is available in very limited areas. It has significant potential because power lines are installed virtually everywhere, alleviating the need to build new broadband facilities for every customer.



Reading comprehension:

1) Complete the paragraph with words from the text:

Broadband Internet service truly is the most used form of Internet access because of its high access; it is offered in different forms, namely DSL (or Digital Subscriber Line), also, cable, and satellite. The old connection is the only non-broadband internet service available, and even though it is cheaper, most users are moving towards the faster broadband Internet connection.

2) Tick the right options:

a. Dial-up connection

- ❖ uses a standard phone line to access to the internet at data transfer rate of up to 56 kbps.
- ❖ is a good opportunity for users who constantly stream videos or play games online.
- ❖ is often the only choice available for rural or remote areas.

b. The broadband connection

- ❖ slows down if more than one user use the internet at the same time.
- ❖ ensures that you have high speed internet at your place.
- ❖ broadband speed differs based on connection type and chosen internet provider.

3) What are the differences between DSL, fiber and cable?

	DSL	Fiber	Cable
Cable type			
Speed			
Availability			

4) State the pros and the cons of satellite connection and BPL.

Satellite connection

+

-

BPL

+

-

5) Find in the last paragraph the synonyms of the following words:

a. capacity:

b. reduce / relieve:

Language:

1. Put the verbs in the right tense or form.

The internet is an immense combination of billions of computers and other connected devices located worldwide and connected via cables and wireless signals. This enormous network **(typify)**..... personal, business, educational and government devices. The internet **(instigate)**..... in the 1960s under the name ARPAnet, as an experiment in how the U.S. military could maintain communications in the case of a possible nuclear strike. With time, ARPAnet became a civic experiment, **(connect)** university mainframe computers for academic purposes. As personal computers became mainstream in the 1980s and 1990s, the internet grew **(exponential)**..... as more users plugged their computers into the **(mass)**

..... network. Today, the internet is growing into a public spiderweb of billions of personal, government, educational and commercial computers and devices; all **(connect)**..... by cables and by wireless signals. No single entity **(own)**the internet. No single government has authority over its **(operate)**..... Some technical rules and hardware and software standards enforce how people plug in to the internet, but for the most part, the internet is a free and open broadcast medium of hardware networking.

3. Complete the paragraph with words from the list:

friendly , radio , sample , wireless , strengthen , microchip , affordable , simultaneously , Wi-Fi

Scientists at Shanghai's Fudan University have created a one-watt light bulb that provides access to wireless internet. Using a technology called Li-Fi, or light fidelity, the prototype created by the scientists is claimed to be an and efficient alternative to existing means. According to Chi Nan, an information technology professor, four computers were found to be able to connect easily to the internet when kept near the Li-Fi bulb. The use of light frequency enables all computers to connect The Li-Fi bulb, featuring a generates around 150 mbps that is 20 times faster than the average connection in China. Nan also states that the current signal transmission equipment is expensive and inefficient." As for cell phones, millions of base stations have been established around the world to the signal but most of the energy is consumed on their cooling systems," Nan told Xinhuanet. The new technology will be showcased at the China International Industry Fair beginning on 5 November in Shanghai, where around 10 - Li-Fi kits will be displayed. The new bulb is seen as a cost-..... product that can greatly help in preventing hacking of wireless signals as light cannot penetrate walls when compared to signals.

4. Fill in the blanks with the suitable prepositions and linkers:

Millions of people living in the U.S. cannot afford the internet, _____ the U.N. has declared it a human right. Nearly one _____ five U.S. households living _____ less than \$30,000 are not online, and half of the non-broadband users say that high costs prevent them _____ subscribing. Even the U.S.'s most economically prosperous states are failing to meet the broadband needs _____ their poorest residents. _____ the 6th largest economy in the world, California is home to nearly 6 million students.

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_____, the average cost _____ broadband _____ home including the router in Los Angeles is \$59.83. But this fails to include activation and installation fees that often exceed \$100. In California, 300,000 students remain without the high-speed internet connection needed to attend classes _____ zoom. Attendance has dropped, largely attributed _____ the fact that over 1.5 million students lacked access _____ reliable internet _____ the start of the pandemic. _____, developing the necessary infrastructure to make high-speed internet available cheaply would cost billions, _____ the costs of failing to address the digital divide and improve internet accessibility would be devastating.

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Writing blog posts

Blogging is about **conversations**; discussing the things that interest you, commenting on events and joining debates.

Common functions for blogging

- Highlight your unique point of view
- Invite comments from your readers
- Respond to comments
- Freely link to relevant blog posts
- Comment on other bloggers' posts

As far as I'm concerned/ with reference to previous posts / I wonder if you share this ? / is this similar to /

I think / propose / strongly believe / admit/ confirm/ object to / I here refer to / hint at (other posts)

useful phrases to express blogging functions

Topic

Developed countries having huge broadband internet infrastructure have already started trying out online education and teleworking. Having heard that, we feel very enthusiastic about it; but is it realizable in our country? "The ball is in your court. Try to work on a 12-line blog post in which you evaluate the broadband infrastructure within your reach.

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